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WORK LIGHT

Abstract

A work light including a base, a light fixture, a magnet, and a boot. The light fixture is supported by the base. The magnet is coupled to the base. The magnet is configured to create a magnetic connection between the base and the work surface to selectively secure the base to the work surface. The boot is coupled to the base and is configured to reduce a strength of the magnetic connection between the base and the work surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/560,929, filed Mar. 4, 2024, the entire contents of which are incorporated by reference herein.

FIELD OF THE INVENTION

[0002] The present disclosure relates to work lights, and more specifically, to work lights configured to be attached to a work surface through magnetism.

SUMMARY

[0003] In one aspect, the disclosure provides a work light including a base, a light fixture, a magnet, and a boot. The light fixture is supported by the base. The magnet is coupled to the base. The magnet is configured to create a magnetic connection between the base and a work surface to selectively secure the base to the work surface. The boot is coupled to the base and is configured to reduce a strength of the magnetic connection between the base and the work surface.

[0004] In some aspects, the boot inhibits the base from being magnetically secured to the work surface.

[0005] In some aspects, the base is securable to the work surface via the magnetic connection while the boot is attached to the base.

[0006] In some aspects, the magnet is exposed on a surface of the base, and the boot covers the magnet.

[0007] In some aspects, the boot is formed of a rubber material.

[0008] In some aspects, the base include an aperture, and the boot includes a protrusion that is configured to be inserted into the aperture to couple the boot to the base. In some aspects the aperture is one of a plurality of apertures in the base, the protrusion is one of a plurality of protrusions on the boot, and each protrusion of the plurality of protrusions is insertable into a corresponding aperture of the plurality of apertures to couple the boot to the base. In some aspects, the protrusion includes a plurality of sweeps, and the base compresses the plurality of sweeps when the protrusion is inserted into the aperture.

[0009] In some aspects, the boot includes a toe portion that is configured to wrap around an end of the base.

[0010] In some aspects, the boot includes a tab that is graspable to assist with removal of the boot from the base.

[0011] In some aspects, the boot includes a magnetic element configured to interact with the magnet to couple the boot to the base.

[0012] In some aspects, the boot includes a strap configured to couple the boot to the base.

[0013] In another aspect, the disclosure provides a work light including a base, a light fixture, a magnet, and a boot. The light fixture is supported by the base. The magnet is coupled to the base and is exposed on a surface of the base. The magnet is configured to create a magnetic connection between the base and a work surface to secure the base to the work surface. The boot is releasably attached to the base and covers the magnet.

[0014] In some aspects, the surface of the base includes an aperture, and the boot includes a protrusion that is configured to be inserted into the aperture to couple the boot to the base. In some aspects, the protrusion includes a plurality of sweeps, and the base compresses the plurality of sweeps when the protrusion is inserted into the aperture.

[0015] In some aspects, the boot includes a toe portion that is configured to wrap around an end of the base.

[0016] In another aspect, the disclosure provides a boot for use with a work light, the work light including a base, a light fixture supported by the base, and a magnet coupled to the base, the

magnet configured to create a magnetic connection between the base and a work surface to selectively secure the base to the work surface. The boot includes a first end, a second end, and a wall. The first end is configured to align with a first end of the base. The second end is configured to align with a second end of the base. The wall extends between the first end and the second end. The wall is configured to reduce a strength of the magnetic connection between the base and the work surface.

[0017] In some aspects, the boot is formed of a rubber material.

[0018] In some aspects, the first end of the boot includes a toe portion that is configured to wrap around the first end of the base.

[0019] In some aspects, the second end of the boot includes a tab that is graspable to assist with removal of the boot from the base.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1A is a perspective view of a work light in a first collapsed state and including a base, an arm, a light fixture, and a manual release assist mechanism according to an embodiment of the disclosure.

[0021] FIG. 1B is a perspective view of the work light in an extended state.

[0022] FIG. 1C is a plan view of the work light in a second collapsed state.

[0023] FIG. 1D is a plan view of the work light in a third collapsed state.

[0024] FIG. 2 is a bottom perspective view of the base of the work light.

[0025] FIG. 3 is an exploded view of the work light, including a boot that is releasably attachable to the base.

[0026] FIG. 4 is a perspective view of the boot of FIG. 3.

[0027] FIG. 5 is a cross-sectional view of a portion of the boot and a portion of the base when the base and the boot are attached.

[0028] FIG. 6 is a perspective of a boot according to another embodiment of the disclosure.

[0029] FIG. 7 is a perspective view of a boot according to another embodiment of the disclosure.

[0030] FIG. 8 is a perspective view of the boot of FIG. 7 attached to the base of the work light of FIG. 1A.

[0031] FIG. 9A is a perspective view of a boot according to another embodiment of the disclosure.

[0032] FIG. 9B is a side view of the boot of FIG. 9A.

[0033] FIG. 10 is a rear view of the boot of FIG. 9A attached to the work light of FIG. 1A.

DETAILED DESCRIPTION

[0034] Before any embodiments of the disclosure are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways.

[0035] FIGS. 1A and 1B illustrate a work light **10**. The work light **10** may also be referred to as a boom light. The boom light **10** is configured to be located on or adjacent to a work surface **5** to provide illumination for a work area. The boom light **10** is adjustable to provide illumination on the work area at different angles and from different heights. The illustrated boom light **10** is just one example of a work light. In other embodiments, the work light may have other configurations and components.

[0036] As illustrated in FIGS. 1A and 1B, the boom light **10** includes a base **14** having a first end **14a** and a second end **14b**, an articulating arm **18**, a light fixture **22**, a handle **26**, and a boot **30** (FIG. 3) that may be releasably attached to the base **14**. The base **14** is configured to be positioned

on or attached to the work surface 5. In the illustrated embodiment, the base 14 is attachable to the work surface 5 through magnetism. In other embodiments, the base 14 may be attachable to the work surface 5 through other means, such as suction or adhesives. In still other embodiments, the base 14 may be attachable to the work surface 5 through a clamp, clips, or other types of fasteners. A battery 34 is removably attached to the base 14 at the first end 14a of the base 14 for providing power to the light fixture 22. The arm 18 is attached to the base 14 between the handle 26 and the battery 34. The arm 18 extends from the base 14 and is attached to the light fixture 22 at an end of the arm 18 opposite from the base 14. The arm 18 is adjustable between a collapsed state (FIG. 1A) and an extended state (FIG. 1B). The handle 26 is integrally formed with the base 14 at the second end 14b of the base 14. In some embodiments, the handle 26 may be separately formed from the base 14 and coupled to the base 14. The handle 26 is configured to assist removal of the base 14 from the work surface 5. The boot 30 may be attached to the base 14 (e.g., at a bottom of the base 14) to, for example, selectively cover the magnets or other attachment means on the base 14.

[0037] As illustrated in FIGS. 1A, 1C, and 1D, the arm 18 may be oriented in a first collapsed state (FIG. 1A), a second collapsed state (FIG. 1C), and a third collapsed state (FIG. 1D). With reference to FIG. 1A, the first collapsed state may also be referred to as an upright collapsed state in which the light fixture 22 extends substantially perpendicular to the base 14. With reference to FIG. 1C, the second collapsed state may also be referred to as a battery-side collapsed state in which the light fixture 22 extends parallel to the base 14 and toward a side of the base 14 (i.e., the first end 14a of the base 14) that receives the battery 34. The arm 18 may only be configurable in the second collapsed state when the battery 34 is not attached to the base 14. With reference to FIG. 1D, the third collapsed state may also be referred to as a handle-side collapsed state in which the light fixture 22 extends parallel to the base 14 and toward a side of the base 14 (i.e., the second end 14b of the base 14) that receives, or includes, the handle 26. As such, the arm 18 enables the boom light 10 to be adaptable to user-preference for, among other things, storing, packing, and transporting the boom light 10. As described in more detail below, the arm 18 is rotatable relative to the base 14 such that the arm 18 may be oriented in other collapsed states than illustrated in FIGS. 1A, 1C, and 1D.

[0038] With reference to FIGS. 1B and 2, the base 14 includes a housing 38, a base hinge 42, and a plurality of magnets 46 disposed within a cavity defined by the housing 38. The housing 38 includes a first portion 38a and a second portion 38b, and the cavity is defined between the first portion 38a and the second portion 38b. In the illustrated embodiment, the first portion 38a may also be referred to as an upper portion, and the second portion 38b may also be referred to as a lower portion. The first portion 38a of the housing 38 defines a battery receptacle 50 (FIG. 1C) that receives the battery 34. The second portion 38b of the housing 38 defines a plurality of apertures 54 that is configured to receive a plurality of fasteners 58 to couple the first portion 38a and the second portion 38b together. The base hinge 42 is positioned on the first portion 38a of the housing 38 and extends, or protrudes, from the first portion 38a of the housing 38. The base hinge 42 is coupled with the articulating arm 18 and enables the arm 18 to rotate relative to the base 14, as will be described in more detail. The plurality of magnets 46 extends from within the cavity through a bottom or magnetic surface 62 of the second portion 38b of the housing 38 such that at least a portion of each of the magnets 46 is exposed from the magnetic surface 62. As such, the plurality of magnets 46 creates a magnetic field for the magnetic surface 62, thereby enabling the magnetic surface 62 to secure the base 14 to the work surface 5 (FIG. 1A) via a magnetic connection with the work surface 5 (FIG. 1A). In the illustrated embodiment, the magnets 46 are permanent block magnets. In other embodiments, the magnets 46 may be formed in a different shape or be a different type of magnet. Additionally or alternatively, the magnets 46 may be positioned on the surface 62 of the base 14, rather than extending through the surface 62 from within the base 14.

[0039] In the illustrated embodiment, the base 14 includes eight magnets 46. In other embodiments, the base 14 may include fewer magnets 46 or more magnets 46. For example, the base 14 may only

include a single magnet. A first group of magnets **46a** is oriented along (e.g., parallel to) a direction extending from the first end **14a** of the base **14** to the second end **14b** of the base **14**. In the illustrated embodiment, the first group of magnets **46a** includes four magnets **46**. A second group of magnets **46b** is oriented perpendicular to the first group of magnets **46a**. In the illustrated embodiment, the second group of magnets **46b** includes four magnets **46**. All of the four magnets **46** from the second group of magnets **46b** are positioned between sets of two magnets **46** from the first group of magnets **46a**.

[0040] With reference to FIGS. **1A** and **1D**, when the battery **34** is received in the battery receptacle **50** (FIG. **1C**), the battery **34** is configured to electrically connect to a printed circuit board assembly (i.e., a PCBA) **63** located in the base **14**. The PCBA **63** is electrically connected to wiring that extends through the arm **18** to the light fixture **22** such that an electrical connection is formed between the light fixture **22** and the battery **34** through the PCBA **63**. Returning reference to FIG. **3**, an actuator or power switch **64** is supported by the housing **38** adjacent to the battery receptacle **50**. The actuator **64** is electrically connected to the PCBA **63** and is actuatable to turn on and off a supply of electricity to the light fixture **22** (FIG. **1A**) from the battery **34**.

[0041] As illustrated in FIGS. **2** and **5**, each of the plurality of apertures **54**, or screw bosses, in the base **14** is substantially circular. In other embodiments, the apertures **54** may have another shape. Each of the apertures **54** has a diameter **DI** sized to receive a corresponding one of the fasteners **58** and a depth **L1** extending from the magnetic surface **62** at least partially into the cavity. In the illustrated embodiment, the plurality of fasteners **58** are screws. Each of the screws **58** may be inserted into a corresponding one of the apertures **54** and tightened with a tool, such as a screwdriver, to secure the second portion **38b** of the housing **38** to the first portion **38a** of the housing **38**, thereby forming the cavity. Specifically, the diameter **DI** of the apertures **54** is large enough to receive the fasteners **58** and small enough to inhibit external forces and/or bodies from loosening the fasteners **58**. The depth **LI** of the apertures **54** allows the fasteners **58** to be recessed from the magnetic surface **62** to further inhibit external forces and/or bodies from loosening the fasteners **58**. In other embodiments, the fasteners **58** may be another type of fastener **58** such as a fastener that may be toollessly tightened and loosened.

[0042] In the illustrated embodiment, the base **14**, and more specifically the second portion **38b** of the housing **38** of the base **14**, defines thirteen apertures **54**. In other embodiments, the base **14** may define fewer apertures **54** or more apertures **54**. The thirteen apertures **54** at least includes three apertures **54** disposed along the first end **14a** of the base **14**, three apertures **54** disposed along the second end **14b** of the base **14**, and two apertures **54** disposed between two sets of magnets **46** from the second group of magnets **46b**.

[0043] Returning reference to FIG. **1B**, the arm **18** includes a first arm segment **66**, a second arm segment **70**, and an arm hinge **74** disposed between the two arm segments **66**, **70** to enable rotation of the arm segments **66**, **70** relative to each other. The first arm segment **66** is coupled to the base hinge **42** at one end of the first arm segment **60**. The base hinge **42** allows the first arm segment **66** to rotate, or pivot, relative to the base **14** about two separate base hinge axes **A1**, **A2**. In the illustrated embodiment, the base hinge axes **A1**, **A2** include a horizontal base hinge axis **A1** that extends skew to the base **14** and a vertical base hinge axis **A2** that extends through the base **14**. The first arm segment **66** is connected to the arm hinge **74** at an end of the first arm segment **66** opposite from the base **14**. The arm hinge **74** connects the first arm segment **66** to the second arm segment **70** and allows the arm segments **66**, **70** to rotate, or pivot, relative to one another about two separate arm hinge axes **A3**, **A4**. In the illustrated embodiment, the arm hinge axes **A3**, **A4** include a horizontal arm hinge axis **A3** that extends parallel to the base hinge horizontal axis **A1** when the arm **18** is in the extended state and a vertical arm hinge axis **A4** that extends coaxial with the base hinge axis **A2** when the arm **18** is in the extended state. Although only two arm segments **66**, **70** and one arm hinge **74** are illustrated in FIG. **1B**, the arm **18** may include additional arm segments **66**, **70** and arm hinges **74** in some embodiments. In other embodiments, the arm **18** may

include just one arm segment **66** and no arm hinges **74**. In still other embodiments, the arm **18** may be omitted entirely. In further embodiments, the arm segments **66**, **70** may be removably couplable to the arm hinges **74** such that the length of the arm **18** is adjustable by adding or removing arm segments **66**, **70** from the arm **18**.

[0044] With reference to FIGS. **1B** and **3**, the light fixture **22** is connected to the arm **18** at an end of the arm **18** opposite from the base **14**. Specifically, the light fixture **22** is pivotably connected to the arm **18** via a light fixture hinge **78** that allows the light fixture **22** to pivot relative to the arm **18** about two separate light fixture axes **A5**, **A6**. In the illustrated embodiment, the light fixture axes **A5**, **A6** include a horizontal light fixture axis **A5** that extends parallel to the horizontal base hinge axis **A1** when the arm **18** is in the extended state and a vertical light fixture axis **A6** that extends coaxial with the vertical base hinge axis **A2** when the arm **18** is in the extended state. The light fixture **22** is electrically coupled to the battery **34** via wiring that extends through the arm **18**. As such, when the boom light **10** is turned on, the battery **34** supplies electricity to the light fixture **22** via the wiring such that the light fixture **22** is configured to emit light through a lens **82**.

Specifically, the light fixture **22** includes a plurality of light emitting diodes (LEDs) that are configured to illuminate when electricity is supplied to the light fixture **22** from the battery **34**. In other embodiments, the light fixture **22** may include other light emitting means. The light fixture **22** further includes a user interface **84** that allows a user to turn the boom light **10** on and off and to adjust the intensity of light emitted through the lens **82** when the boom light **10** is on.

[0045] With reference to FIGS. **2-4**, the boot **30** is shaped complementarily to the lower portion **38b** of the housing **38** of the base **14** such that when the boot **30** is coupled to the base **14**, the boot **30** substantially encapsulates the magnetic surface **62**. As such, the boot **30** includes a first end **30a** that is configured to align with the first end **14a** of the base **14** and a second end **30b** that is configured to align with the second end **14b** of the base **14**. The illustrated boot **30** also includes a wall **30c** extending between the first end **30a** and the second end **30b**. The wall **30c** may also be referred to as a bottom wall. The wall **30c** is generally planar and shaped and sized to cover the magnetic surface **62** of the base **14**. In the illustrated embodiment, the wall **30c** covers the entire magnetic surface **62**. In other embodiments, the wall **30c** may cover only a portion of the magnetic surface **62**. In the illustrated embodiment, the boot **30** is formed of a rubber material. The rubber material may be, for example, a natural rubber, a butyl rubber, a nitrile rubber, or another type of rubber. The rubber material of the boot **30** may inhibit wear caused by, among other things, engagement between the magnetic surface **62** and the work surface **5** (FIG. **1A**), thereby increasing the lifespan of the base **14** of the boom light **10**. The boot **30** may also help protect the work surface **5** from marring or scratches. Additionally, the boot **30** may adjust the strength of the magnetic connection between the magnetic surface **62** and the work surface **5** or other objects. Specifically, the material of the boot **30** provides a medium between the magnetic surface **62** and the work surface **5** (FIG. **1A**) that increases the distance between the magnets **46** and the work surface **5** (FIG. **1A**), thereby reducing the strength of the magnetic connection between the base **14** and the work surface **5** (FIG. **1A**). However, the base **14** may still be securable to the work surface **5** (FIG. **1A**) via magnetic connection with the boot **30** attached to the base **14**. As such, the boot **30** may reduce the amount of force that is required to overcome the magnetic connection between the base **14** and the work surface **5** (FIG. **1A**) to remove the base **14** from the work surface **5** (FIG. **1A**). In other scenarios, such as when the boom light **10** is being stored in a toolbox or other container, the boot **30** may inhibit the magnets **46** from interacting with other objects in the toolbox or other container. In some embodiments, the boot **14** may be thin enough and/or made of a suitable material to not alter the magnetic connection between the base **14** and the work surface **5**.

[0046] As illustrated in FIGS. **4** and **5**, the boot **30** includes a coupling interface **85** that is configured to engage the magnetic surface **62** of the base **14** (FIG. **2**) to releasably attach the boot **30** to the base **14**. The coupling interface **85** has a plurality of protrusions **86** for coupling the boot **30** to the base **14**. Each protrusion **86** has a cylindrical body **86a** such that the protrusions **86** are

shaped to be inserted into the circular apertures **54** of the second portion **38b** of the housing **38** (FIG. 1A) to couple the boot **30** to the base **14**. In the illustrated embodiment, the plurality of protrusions **86** includes eight protrusions **86**. Specifically, the plurality of protrusions **86** includes three protrusions **86** disposed along the first end **30a** of the boot **30** that are configured to be inserted into the three apertures **54** disposed along the first end **14a** of the base **14**, three protrusions **86** disposed along the second end **30b** of the boot **30** that are configured to be inserted into the three apertures **54** disposed along the second end **14b** of the base **14**, and two protrusions **86** disposed roughly in the middle of the boot **30** (e.g., between the first end **30a** and the second end **30b**) that are configured to be inserted into the apertures **54** disposed between the two sets of magnets **46** from the second group of magnets **46**. Each of the protrusions **86** includes a plurality of sweeps **90** equally spaced from each other around the cylindrical body **86a** and extending outward from the cylindrical body **86a** of the protrusion **86**. In the illustrated embodiment, the sweeps **90** are compressible. When the protrusions **86** are inserted into the apertures **54** to couple the boot **30** to the base **14**, the base **14** compresses the sweeps **90**. The compression of the sweeps **90** creates a friction fit between the protrusions **86** of the boot **30** and the second portion **38b** of the housing **38** (FIG. 1A) of the base **14**. As such, the friction fit between the protrusions **86** of the boot **30** and the base **14** inhibits the boot **30** from falling off of the base **14**. Specifically, the friction fit inhibits the boot **30** from being removed from the base **14** unless a force is applied to either the base **14** or the boot **30** that overcomes the friction fit.

[0047] FIG. 6 illustrates another embodiment of a boot **130** for the boom light **10** of FIG. 1A. With reference to FIGS. 1A and 6, the boot **130** is shaped complementarily to the second portion **38b** of the housing **38** of the base **14** such that when the boot **130** is coupled to the base **14**, the boot **130** substantially encapsulates the magnetic surface **62** (FIG. 2). As such, the boot **130** includes a first end **130a** that is configured to align with the first end **14a** of the base **14** and a second end **130b** that is configured to align with the second end **14b** of the base **14**. The illustrated boot **130** also includes a wall **130c** extending between the first end **130a** and the second end **130b**. The wall **130c** may also be referred to as a bottom wall. The wall **130c** is generally planar and shaped and sized to cover the magnetic surface **62** of the base **14**. In the illustrated embodiment, the wall **130c** covers the entire magnetic surface **62**. In other embodiments, the wall **130c** may cover only a portion of the magnetic surface **62**. The boot **130** is formed of a hard plastic material and may be formed through an overmolding process. The hard plastic material may be, for example, high density polyethylene (HDPE), a polyether ether ketone (PEEK), or another similar material. The boot **130** may be inhibit wear caused by, among other things, engagement between the magnetic surface **62** (FIG. 2) and the work surface **5**, thereby increasing the lifespan of the boom light **10**. The boot **130** may also help protect the work surface **5** from marring or scratches. Additionally, the boot **130** is relatively thicker than the boot **30** of FIG. 4. As such, the boot **130** has a thickness that increases the distance between the magnets **46** (FIG. 2) and the work surface **5** to a distance in which the strength of the magnetic connection between the magnetic surface **62** (FIG. 2) and the work surface **5** is reduced and the base **14** of the boom light **10** does not magnetically attach to the work surface **5**.

[0048] The boot **130** includes a plurality of receptacle slots **132** and a plurality of metal strips **134** positioned in the plurality of receptacle slots **132** for coupling the boot **130** to the base **14**. In some embodiments, the metal strips **134** may be insert molded in the receptacle slots **132**. In the illustrated embodiment, the boot **130** includes eight receptacle slots **132** and four metal strips **134**. Each of the receptacle slots **132** is formed to align with a corresponding one of the magnets **46** in the base **14** (FIG. 2). As such, the eight receptacle slots **132** include a first group of receptacle slots **132a** that extends along (e.g., parallel to) a direction extending from the first end **130a** of the boot **130** to the second end **130b** of the boot **130** and a second group of receptacle slots **132b** that extends perpendicular to the first group of receptacle slots **132a**. In the illustrated embodiment, each of the four metal strips **134** is supported in a corresponding one of the receptacle slots **132**

from the first group of receptacle slots **132a**. In some embodiments, each of the four metal strips **134** may be supported in a corresponding one of the receptacle slots **132** from the second group of receptacle slots **132b**. In other embodiments, the four metal strips **134** may be supported in any combination of receptacle slots **132** from the first group of receptacle slots **132a** and the second group of receptacle slots **132b**. In further embodiments, the boot **130** may include eight metal strips **134** such that a metal strip **134** is supported in each of the receptacle slots **132**. The plurality of magnets **46** interact with the metal strips **134**, thereby enabling the boot **130** to secure to the base **14** via a magnetic connection between the magnets **46** and the metal strips **134**. As such, the magnetic connection between the magnets **46** and the metal strips **134** inhibits the boot **130** from being removed from the base **14** unless a force is applied to either the base **14** or the boot **130** that overcomes the magnetic connection. The magnetic connection between the magnets **46** and the metal strips **134** may additionally reduce the magnetic connection between the base **14** (FIG. 2) and the work surface **5** (FIG. 1A) such that the boot **130** inhibits the base **14** from being magnetically secured to the work surface **5**. In some embodiments, the metal strips **134** may be replaced with magnets that interact with the magnets **46** on the base **14**. In still other embodiments, the boot **130** may include other types of magnetic elements that interact with the magnets **46** on the base **14**. The boot **130** further includes a tab **138** that protrudes from the second end **130b** of the boot **130** and that is graspable to assist a user with pulling the boot **130** off of the base **14**.

[0049] FIGS. 7 and 8 illustrate another embodiment of a boot **230** for the boom light **10** of FIG. 1A. With reference to FIGS. 1A, 7, and 8, the boot **230** is shaped complementarily to the second portion **38b** of the housing **38** of the base **14** such that when the boot **230** is coupled to the base **14**, the boot **230** substantially encapsulates the magnetic surface **62** (FIG. 2). As such, the boot **230** includes a first end **230a** that is configured to align with the first end **14a** of the base **14** and a second end **230b** that is configured to align with the second end **14b** of the base **14**. The boot **230** further includes a first side **230c** that extends between the first end **230a** and the second end **230b** and second side **230d** that extends between the first end **230a** and the second end **230b** opposite from the first side **230c**. The illustrated boot **230** also includes a wall **230e** extending between the first end **230a** and the second end **230b**. The wall **230e** may also be referred to as a bottom wall. The wall **230e** is generally planar and shaped and sized to cover the magnetic surface **62** of the base **14**. In the illustrated embodiment, the wall **230e** covers the entire magnetic surface **62**. In other embodiments, the wall **230e** may cover only a portion of the magnetic surface **62**. The boot **230** is formed of substantially the same material as the boot **30** of FIG. 4. Therefore, the boot **230** may inhibit wear caused by, among other things, engagement between the magnetic surface **62** (FIG. 2) and the work surface **5**, thereby increasing the lifespan of the base **14** of the boom light **10**. The boot **230** may also help protect the work surface **5** from marring or scratches. Additionally, the boot **230** may adjust the strength of the magnetic connection between the magnetic surface **62** (FIG. 2) and the work surface **5** such that the boot **230** reduces the amount of force that is required to overcome the magnetic connection between the base **14** and the work surface **5** to remove the base **14** from the work surface **5**.

[0050] The boot **230** includes a toe portion **234**, a strap **238**, and a clasp **242** for coupling the boot **230** to the base **14**. The toe portion **234** is provided at the first end **230a** of the boot **230** and is configured to wrap around the first end **14a** of the base **14** when the boot **230** is coupled to the base **14**. Stated another way, the toe portion **234** of the boot **230** substantially encapsulates the first end **14a** of the base **14** when the boot **230** is coupled to the base **14**. The strap **238** is positioned roughly in the middle of the boot **230** and includes a first end **238a** positioned at the first side **230c** of the boot **230** and a second end **238b** opposite the first end **238a**. The clasp **242** is positioned at the second side **230d** of the boot **230**. The strap **238** is adjustable between an open configuration and a closed configuration to allow the boot **230** to be selectively attached and removed from the base **14**. Specifically, the strap **238** includes strap apertures **246a**, **246b**, **246c** at the second end **238b** of the strap **238** that are configured to selectively receive the clasp **242** to adjust the strap between the

open configuration and the closed configuration. That is, when the clasp **242** is received in one of the strap apertures **246a-246c**, the clasp **242** holds the strap **238** in the closed configuration. The clasp **242** may be received in any one of the apertures **246a-246c** to adjust the tightness with which the strap **238** secures the boot **230** to the base **14**. In other embodiments, the second end **238b** of the strap **238** may be secured to the boot **230** by other mechanisms. For example, the strap **238** may be secured by hook and loop fasteners (e.g., Velcro®), snaps, buttons, a buckle, a ratchet-strap mechanism, and the like.

[0051] As shown in FIG. **8**, when the boot **230** is coupled to the base **14** of the boom light **10**, the strap **238** wraps around a portion of the base **14**. In the illustrated embodiment, the strap **238** is positioned between the arm **18** and the battery receptacle **50** on the base **14**. Such an arrangement helps protect the strap **14** and inhibits the strap **238** from shifting or sliding on the base **14**. In other embodiments, the strap **14** may wrap around other portions of the base **14** or the boom light **10**.

[0052] FIGS. **9A-10** illustrate another embodiment of a boot **330** for the boom light **10** of FIG. **1A**. With reference to FIGS. **1**, **9A**, and **9B**, the boot **330** is shaped complementarily to the second portion **38b** of the housing **38** of the base **14** such that when the boot **330** is coupled to the base **14**, the boot **330** substantially encapsulates the magnetic surface **62** (FIG. **2**). As such, the boot **330** includes a first end **330a** that is configured to align with the first end **14a** of the base **14** and a second end **330b** that is configured to align with the second end **14b** of the base **14**. The illustrated boot **330** also includes a wall **330c** extending between the first end **330a** and the second end **330b**. The wall **330c** may also be referred to as a bottom wall. The wall **330c** is generally planar and shaped and sized to cover the magnetic surface **62** of the base **14**. In the illustrated embodiment, the wall **330c** covers the entire magnetic surface **62**. In other embodiments, the wall **330c** may cover only a portion of the magnetic surface **62**. The boot **330** is formed of substantially the same material as the boot **30** of FIG. **4**. Therefore, the boot **330** may inhibit wear caused by, among other things, engagement between the magnetic surface **62** (FIG. **2**) and the work surface **5**, thereby increasing the lifespan of the base **14** of the boom light **10**. The boot **330** may also help protect the work surface **5** (FIG. **1A**) from marring or scratches. Additionally, the boot **330** may adjust the strength of the magnetic connection between the magnetic surface **62** (FIG. **2**) and the work surface **5** such that the boot **330** reduces the amount of force that is required to overcome the magnetic connection between the base **14** and the work surface **5** to remove the base **14** from the work surface **5**.

[0053] The boot **330** includes a coupling interface **332** that is configured to engage the magnetic surface **62** of the base **14** (FIG. **2**) to releasably attach the boot **330** to the base **14**. Specifically, the coupling interface **332** includes a plurality of protrusions **334** and a toe portion **338** for coupling the boot **330** to the base **14**. The boot **330** additionally includes a tab **342** to assist with removal of the boot **330** from the base **14**. The protrusions **334** are substantially similar to the protrusions **86** of FIG. **4**, and the toe portion **338** is substantially similar to the toe portion **234** of FIG. **7**. As such, each protrusion **334** has a cylindrical body **334a** that enables the protrusions **334** to be inserted into the circular apertures **54** of the second portion **38b** of the housing **38** (FIG. **1A**) to assist with coupling the boot **330** to the base **14**. In the illustrated embodiment, the plurality of protrusions **334** includes three protrusions **334** disposed along the second end **334b** of the boot **330** that are configured to be inserted into the three apertures **54** disposed along the first end **14a** of the base **14**. Each of the protrusions **334** includes a plurality of sweeps **346** equally spaced from each other around the cylindrical body **334a** and extending outward from the cylindrical body **334a** of the protrusion **334**. In the illustrated embodiment, the sweeps **346** are compressible to create a friction fit between the protrusions **334** and the housing **38** of the base **14** (FIG. **1A**), as similarly described with respect to the protrusions **86** and sweeps **90** of FIG. **4**.

[0054] The toe portion **338** is provided at the first end **330a** of the boot **330** and is configured to wrap around the first end **14a** of the base **14** when the boot **330** is coupled to the base **14**. Stated another way, the toe portion **338** of the boot **330** substantially encapsulates the first end **14a** of the

base **14** when the boot is coupled to the base **14**. As such, the protrusions **334** and the boot **330** assist with coupling the boot **330** to the base **14** and inhibiting the boot **330** from being removed from the base **14**. The tab **342** is provided at the second end **330b** of the boot **330** and protrudes from a rear wall **348** of the boot **330**. When the boot **330** is coupled to the base **14**, as illustrated in FIG. **10**, the tab **342** is positioned adjacent to the handle **26**. As such, the tab **342** is configured to be grasped to assist with removal of the boot **330** from the base **14**. For example, a user may grasp the handle **26** with one hand and the tab **342** with the other hand and pull the handle **26** and the tab **342** in opposite directions to remove the boot **330** from the base **14**.

[0055] As illustrated in FIG. **9B**, the boot **330** further includes indicia **350** on a side of the boot **330**. The indicia **350** designate a magnetic status of the base **14**. Specifically, a first indicium **350a** includes a shield and an inactive magnet to designate that the boot **330** substantially inhibits or reduces magnetic engagement for the base **14** while the boot **330** is coupled to the base **14**. A second indicium **350b** includes an active magnet and no shield to designate that the base **14** is able to magnetically engage other objects with the boot **330** removed from the base **14**.

[0056] Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described.

Claims

1. A work light comprising: a base; a light fixture supported by the base; a magnet coupled to the base, the magnet configured to create a magnetic connection between the base and a work surface to selectively secure the base to the work surface; and a boot coupled to the base and configured to reduce a strength of the magnetic connection between the base and the work surface.
2. The work light of claim 1, wherein the boot inhibits the base from being magnetically secured to the work surface.
3. The work light of claim 1, wherein the base is securable to the work surface via the magnetic connection while the boot is attached to the base.
4. The work light of claim 1, wherein the magnet is exposed on a surface of the base, and wherein the boot covers the magnet.
5. The work light of claim 1, wherein the boot is formed of a rubber material.
6. The work light of claim 1, wherein the base includes an aperture, and wherein the boot includes a protrusion that is configured to be inserted into the aperture to couple the boot to the base.
7. The work light of claim 6, wherein the aperture is one of a plurality of apertures in the base, wherein the protrusion is one of a plurality of protrusions on the boot, and wherein each protrusion of the plurality of protrusions is insertable into a corresponding aperture of the plurality of apertures to couple the boot to the base.
8. The work light of claim 6, wherein the protrusion includes a plurality of sweeps, and wherein the base compresses the plurality of sweeps when the protrusion is inserted into the aperture.
9. The work light of claim 1, wherein the boot includes a toe portion that is configured to wrap around an end of the base.
10. The work light of claim 1, wherein the boot includes a tab that is graspable to assist with removal of the boot from the base.
11. The work light of claim 1, wherein the boot includes a magnetic element configured to interact with the magnet to couple the boot to the base.
12. The work light of claim 1, wherein the boot includes a strap configured to couple the boot to the base.
13. A work light comprising: a base; a light fixture supported by the base; a magnet coupled to the base and exposed on a surface of the base, the magnet configured to create a magnetic connection between the base and a work surface to secure the base to the work surface; and a boot releasably

attached to the base and covering the magnet.

14. The work light of claim 13, wherein the surface of the base includes an aperture, and wherein the boot includes a protrusion that is configured to be inserted into the aperture to couple the boot to the base.

15. The work light of claim 14, wherein the protrusion includes a plurality of sweeps, and wherein the base compresses the plurality of sweeps when the protrusion is inserted into the aperture.

16. The work light of claim 13, wherein the boot includes a toe portion that is configured to wrap around an end of the base.

17. A boot for use with a work light, the work light including a base, a light fixture supported by the base, and a magnet coupled to the base, the magnet configured to create a magnetic connection between the base and a work surface to selectively secure the base to the work surface, the boot comprising: a first end configured to align with a first end of the base; a second end configured to align with a second end of the base; and a wall extending between the first end and the second end, the wall configured to reduce a strength of the magnetic connection between the base and the work surface.

18. The boot of claim 17, wherein the boot is formed of a rubber material.

19. The boot of claim 17, wherein the first end of the boot includes a toe portion that is configured to wrap around the first end of the base.

20. The boot of claim 17, wherein the second end of the boot includes a tab that is graspable to assist with removal of the boot from the base.
